Coasters and pendulums

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use height differences to predict speed in ideal motion.

SUCCESS CRITERIA - I can...

1. Identify the relevant vertical height change.
2. Conserve energy under stated assumptions.
3. Separate path length from gravitational energy change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

Gravitational potential energy changes from -15 J to -3 J. Find ΔU and state whether energy is gained or lost.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $mg\Delta h = \frac{1}{2}mv^2$; $v = \sqrt{(2\ g\Delta h)}$.
Point mass released from rest, no losses or rotation; Δh is the positive vertical drop.

Worked reasoning: A pendulum released from rest 0.8 m above its lowest point reaches speed $\sqrt{(2\ gh)}=4$ m/s there with g=10 and negligible losses.

Your turn: Sketch the path and energy bars at the release, bottom and opposite turning point.

Marzano check - circle: ① ② ③

03 Look and explain

Two frictionless tracks start and end at the same heights but take different paths. Compare final speeds from rest.

Marzano check - circle: ① ② ③

04 Build the solution

A 2 kg cart drops 3 m. Use g=10. Find lost potential energy and gained kinetic energy.

Marzano check - circle: ① ② ③

05 Use the idea

A classmate uses total ramp length as h in mgh. Explain how to identify the correct measurement.

Marzano check - circle: ① ② ③

06 Choose a method

A fixed impulse changes a target's sideways velocity by 0.04 m/s. Find the modeled shift after 600 s and state a limitation.

Marzano check - circle: ① ② ③

07 Exit check

A pendulum loses 12 J of gravitational energy while moving down. Neglect losses. What is its kinetic energy gain?

Marzano check - circle: ① ② ③

08 STRETCH: Apply it another way

A pendulum returns to only 0.6 m above the bottom. For a 1 kg bob with g=10, find energy transferred out of its mechanical-energy account.

Marzano check - circle: ① ② ③